

Unless stated otherwise, each question is of equal value [4 pts]. Show all of your work.

Find the domain and range of the following function.

1. $g(x, y) = \sqrt{x - 2y}$

Domain: $\{(x, y) : x - 2y \geq 0\}$

Range: $[0, \infty)$

Evaluate the following limits.

2. $\lim_{(x,y) \rightarrow (1,0)} (4x - y)$

$= 4(1) - 0 = 4$ since the function is continuous on $\mathbb{R}^2$.

3. $\lim_{(x,y) \rightarrow (1,-2)} \frac{y^2 + 2xy}{y + 2x}$

$\lim_{(x,y) \rightarrow (1,-2)} \frac{\cancel{yy} + \cancel{2x}}{\cancel{y} + \cancel{2x}} = \lim_{(x,y) \rightarrow (1,-2)} y = -2$

Use the Two-Path Test to prove that the following limit do not exist.

4. $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$

Take the path $y = x$, then $\lim_{(x,x) \rightarrow (0,0)} \frac{x^2}{2x^2} = \frac{1}{2}$.

But if we take the path $y = -x$, then $\lim_{(x,-x) \rightarrow (0,0)} \frac{-x^2}{x^2 + (-x)^2} = \lim_{(x,-x) \rightarrow (0,0)} \frac{-x^2}{2x^2} = -\frac{1}{2}$. So

$\lim_{(x,-x) \rightarrow (0,0)} \frac{-x^2}{x^2 + (-x)^2}$ DNE.

Find the indicated partial derivative of the following functions.

5. Using $h(x, y) = y^3 \sin 4x$, find h_{xx} and h_{xy} .

$h_x = 4y^3 \cos 4x$

$h_{xx} = -16y^3 \sin 4x$ and $h_{xy} = 12y^2 \cos 4x$.

Use the Chain Rule to find the following derivative. When feasible express your answer in terms of the independent variable.

6. [6 pts] dz/dt where $z = x^3y - x$, $x = t^2$, and $y = 2t^{-1}$.

$\frac{dz}{dx} = 3x^2y - 1$, $\frac{dx}{dt} = 2t$, $\frac{dz}{dy} = x^3$ and $\frac{dy}{dt} = -2t^{-2}$. $\frac{dz}{dt} = \frac{dz}{dx} \frac{dx}{dt} + \frac{dz}{dy} \frac{dy}{dt}$
So

$$\begin{aligned} \frac{dz}{dt} &= (3x^2y - 1)(2t) + (x^3)(-2t^{-2}) \\ &= 2t(3t^4 2t^{-1} - 1) - 2t^{-2}t^6 \\ &= 12t^4 - 2t - 2t^4 = 10t^4 - 2t \end{aligned}$$

Use the function $f(x, y) = x^2 - 4y^2 - 9$ to complete problems 7–9.

7. Compute the gradient of $f(x, y)$ at $P(1, -2)$.

$$f_x = 2x \text{ and } f_y = -8y.$$

$$\nabla f(1, -2) = \langle 2x, -8y \rangle|_{(1, -2)} = \langle 2, 16 \rangle$$

8. Find vectors that give the direction of steepest ascent and descent for f at $P(1, -2)$.

$$\text{ascent: } \nabla f(1, -2) = \langle 2, 16 \rangle \text{ and descent: } -\nabla f(1, -2) = -\langle 2, 16 \rangle$$

9. Compute the directional derivative of $f(x, y)$ at $P(1, -2)$ in the direction of $\mathbf{u} = \left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$ (note that $|\mathbf{u}| = 1$).

$$\left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle \cdot \langle 2, 16 \rangle = \frac{2}{\sqrt{5}} + \frac{32}{\sqrt{5}} = \frac{34}{\sqrt{5}}$$

Use the Second Derivative Test to determine (if possible) whether each critical point of the given function is a local maximum, local minimum, or saddle point.

10. $f(x, y) = 4 + x^3 + y^3 - 3xy$ with critical points:

$$f_x = 3x^2 - 3y, \quad f_{xx} = 6x, \quad f_y = 3y^2 - 3x, \quad \text{saddle point.}$$

$$f_{yy} = 6y, \text{ and } f_{xy} = -3$$

$$D_f(x, y) = 36xy - 9$$

- a. $(0, 0) : D_f(0, 0) = 0 \cdot 0 - 9 < 0 \Rightarrow (0, 0)$ is a saddle point.
- b. $(1, 1) : D_f(1, 1) = 36 - 9 > 0$ and $f_{xx}(1, 1) = 6 > 0 \Rightarrow (1, 1)$ is a local minimum.

Find all of the critical points (there are two) of the following function.

11. $f(x, y) = \frac{x^3}{3} - \frac{y^3}{3} + 3xy$

$$f_x = x^2 + 3y \text{ and } f_y = -y^2 + 3x$$

$$x^2 + 3y = 0 \Rightarrow x = 0 \text{ and } y = 0 \Rightarrow (0, 0) \text{ OR } x^2 = -3y \Rightarrow y = -x^2/3 \Rightarrow x^4 - 27x = 0$$

$$\Rightarrow x(x^3 - 27) = 0 \Rightarrow x = 0 \text{ or } x = 3 \text{ thus } (3, -3) \text{ and } (0, 0) \text{ are critical points.}$$